

TS1: Síntesis y operaciones de señales

Ej 2: $h[n] = \delta[n] - \delta[n-4]$

$x[n] = \cos(\omega_0 n T_s)$

$y[n] = \cos(\omega_0 n T_s) * h[n] \Rightarrow y[n] = \sum_{k=-\infty}^{\infty} \cos(\omega_0 k T_s) \cdot h[n-k]$

$= \sum_{k=-\infty}^{\infty} \cos(\omega_0 k T_s) \cdot (\delta[n-k] - \delta[n-4-k]) =$

$= \sum_{k=-\infty}^{\infty} \cos(\omega_0 k T_s) \cdot \delta[n-k] - \sum_{k=-\infty}^{\infty} \cos(\omega_0 k T_s) \cdot \delta[n-4-k]$

Seleccionan ciertos términos

no nulo cuando $n=k$

no nulo cuando $n=-4-k$

$\Rightarrow y[n] = \cos(\omega_0 n T_s) - \cos(\omega_0 n T_s - \omega_0 4 T_s)$

$y[n] = \frac{e^{j\omega_0 n T_s} + e^{-j\omega_0 n T_s}}{2} - \left(\frac{e^{j\omega_0 n T_s} e^{-j\omega_0 4 T_s} + e^{-j\omega_0 n T_s} e^{j\omega_0 4 T_s}}{2} \right)$

$y[n] = \frac{e^{j\omega_0 n T_s}}{2} (1 - e^{-j\omega_0 4 T_s}) + \frac{e^{-j\omega_0 n T_s}}{2} (1 - e^{j\omega_0 4 T_s})$

Utilizo propiedades trigonométricas para descomponer los términos. Por ejemplo:

$1 - e^{-j\omega_0 4 T_s} = e^{-j\omega_0 2 T_s} (e^{j\omega_0 2 T_s} - e^{-j\omega_0 2 T_s})$

$\hookrightarrow 1 - e^{-j\omega_0 4 T_s} = e^{-j\omega_0 2 T_s} \cdot 2j \operatorname{sen}(2\omega_0 T_s) \rightarrow j = e^{j\frac{\pi}{2}}$

$\Rightarrow \frac{1}{2}(1 - e^{-j\omega_0 4 T_s}) = \operatorname{sen}(2\omega_0 T_s) \cdot e^{j\frac{\pi}{2}} \cdot e^{-j\omega_0 2 T_s} = \operatorname{sen}(2\omega_0 T_s) e^{j(\frac{\pi}{2} - \omega_0 2 T_s)}$

Aplicando la misma idea:

$1 - e^{j\omega_0 4 T_s} = e^{j\omega_0 2 T_s} (e^{-j\omega_0 2 T_s} - e^{j\omega_0 2 T_s}) = -2j \operatorname{sen}(2\omega_0 T_s)$

$-j = e^{-j\frac{\pi}{2}}$

$\Rightarrow 1 - e^{j\omega_0 4 T_s} = -2j \operatorname{sen}(2\omega_0 T_s) \cdot e^{j\omega_0 2 T_s} = 2 \operatorname{sen}(2\omega_0 T_s) \cdot e^{j(\omega_0 2 T_s - \frac{\pi}{2})}$

$\frac{1 - e^{j\omega_0 4 T_s}}{2} = \operatorname{sen}(2\omega_0 T_s) \cdot e^{j(\omega_0 2 T_s - \frac{\pi}{2})}$

$\Rightarrow y[n] = e^{j\omega_0 n T_s} \cdot \operatorname{sen}(2\omega_0 T_s) \cdot e^{j(\frac{\pi}{2} - \omega_0 2 T_s)} + e^{-j\omega_0 n T_s} \cdot \operatorname{sen}(2\omega_0 T_s) \cdot e^{j(\omega_0 2 T_s - \frac{\pi}{2})}$

$$y[n] = \sin(2\omega_0 T_s) \left[e^{j(\omega_0 T_s n + \pi/2 - \omega_0 T_s 2)} + e^{-j(\omega_0 T_s n + \pi/2 - \omega_0 T_s 2)} \right]$$

$$2 \cos(\omega_0 T_s n + \pi/2 - \omega_0 T_s 2)$$

$$\Rightarrow y[n] = \underbrace{2}_{A} \cdot \underbrace{\sin(2\omega_0 T_s)}_{\phi} \cdot \underbrace{\cos(\omega_0 T_s n + \pi/2 - \omega_0 T_s 2)}_{\cos(\alpha)}$$

$$b - x[n] = \left(\frac{1}{2}\right)^n u[n]$$

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} \left(\frac{1}{2}\right)^k u[k] \cdot \delta[n-k] - \sum_{k=-\infty}^{\infty} \left(\frac{1}{2}\right)^k u[k] \cdot \delta[n-4-k]$$

$$y[n] = \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k \delta[n-k] - \sum_{k=4}^{\infty} \left(\frac{1}{2}\right)^k \delta[n-4-k] =$$

$$y[n] = \left(\frac{1}{2}\right)^n u[n] - \left(\frac{1}{2}\right)^{n-4} u[n-4]$$

se mostrarán valores en que $k=n-4$

se mostrarán valores en que $k=n$

$$c - x[n] = u[n+1] - u[n-2]$$

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} (u[k+1] - u[k-2]) (\delta[n-k] - \delta[n-4-k]) =$$

$$y[n] = \sum_{k=-\infty}^{\infty} u[k+1] \cdot \delta[n-k] - u[k+1] \cdot \delta[n-4-k] - u[k-2] \delta[n-k] + u[k-2] \cdot \delta[n-4-k]$$

$$y[n] = u[n+1] - u[n-3] - u[n-2] + u[n-6]$$

$$N=8$$

$$\Rightarrow y[n] = \left(\frac{1}{2}\right)^n \cdot u[n] - \left(\frac{1}{2}\right)^{n-4} u[n-4]$$

$$y[0] = \left(\frac{1}{2}\right)^0 \cdot 1 = 1 \quad y[2] = \left(\frac{1}{2}\right)^2 \cdot 1 = 1/4$$

$$y[1] = \left(\frac{1}{2}\right)^1 \cdot 1 = 1/2 \quad y[3] = \left(\frac{1}{2}\right)^3 \cdot 1 = 1/8$$

$$y[4] = \left(\frac{1}{2}\right)^4 \cdot 1 - \left(\frac{1}{2}\right)^0 \cdot 1 = 1/16 - 1 = -15/16$$

$$y[5] = \left(\frac{1}{2}\right)^5 \cdot 1 - \left(\frac{1}{2}\right)^1 \cdot 1 = 1/32 - 1/2 = -15/32$$

$$y[6] = \left(\frac{1}{2}\right)^6 \cdot 1 - \left(\frac{1}{2}\right)^2 \cdot 1 = 1/64 - 1/4 = -15/64$$

$$y[7] = \left(\frac{1}{2}\right)^7 \cdot 1 - \left(\frac{1}{2}\right)^3 \cdot 1 = 1/128 - 1/8 = -15/128$$

$$y[8] = \left(\frac{1}{2}\right)^8 \cdot 1 - \left(\frac{1}{2}\right)^4 \cdot 1 = 1/256 - 1/16 = -15/256$$

$$c-x[n] = u[n+1] - u[n-3] - u[n-2] + u[n-6]$$

$$y[-1] = u[0] = 1 \quad y[0] = u[1] = 1$$

$$y[1] = u[2] = 1 \quad y[2] = u[3] - u[0] = 0$$

$$y[3] = u[4] - u[0] - u[1] = -1$$

$$y[4] = u[5] - u[1] - u[2] = -1$$

$$y[5] = u[6] - u[2] - u[3] = -1$$

$$y[6] = u[7] - u[3] - u[4] + u[0] = 0$$

$$y[7] = u[8] - u[4] - u[5] + u[1] = 0$$

$$y[8] = u[9] - u[5] - u[6] + u[2] = 0$$